

Eduardo Figueroa

COMPUTER SCIENCE STUDENT · BIOINFORMATICS RESEARCHER · DATA SCIENCE ENTHUSIAST

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Professional Summary

Undergraduate Computer Science student actively seeking research experiences or graduate schools in Bioinformatics.

Education

University of Puerto Rico, Rio Piedras

San Juan, Puerto Rico

BACHELOR OF SCIENCE IN COMPUTER SCIENCE, MINOR IN CYBERSECURITY

Expected December 2026

- **Relevant Courses:** Cybersecurity, Networking, Data Science, Bioinformatics, Operating Systems, Intro to Computational Statistics, API Development

Professional Experience

Bioinformatics Assistant

San Juan, Puerto Rico

GENOS, LLC

January 2026 - Present

Research Assistant - COVID-19 Diabetes Study

San Juan, Puerto Rico

ABARTYS (WORK INTERNSHIP)

May 2025 - July 2025

- Organized data and created patient cohorts for COVID-19 diabetes research
- Implemented machine learning models for data analysis
- Conducted feature importance analysis using local datasets
- **Mentors:** Dr. Roche-Lima, Dr. Velev

Research Experience

FAIR Metrics Development Researcher

Pittsburgh, Pennsylvania

CARNEGIE MELLON UNIVERSITY

January 2025 - November 2025

- Developed, assessed and visualized FAIR metrics for datasets in HuBMAP
- Created custom metrics for each HuBMAP dataset
- Managed data evolution and new schemas
- Designed interface to select views of FAIRness data
- **Mentor:** I. Cao-Berg

Internal Tools Developer

San Juan, Puerto Rico

UPR MEDICAL SCIENCE

November 2025 - December 2025

- Evaluated and modified internal tools for data processing
- Designed user-friendly interfaces for data management
- **Mentors:** Brenda Nieves, Dr. Roche-Lima

HuBMAP Dashboard Developer

Pittsburgh, Pennsylvania

CARNEGIE MELLON UNIVERSITY

May 2024 - August 2024

- Designed metrics for HuBMAP dashboard
- Created different views for stakeholders (data curators, public view, PI)
- Implemented automatic FAIRness general metrics using Python and Grafana
- Used Bridges 2 for processing computational analysis
- **Mentor:** I. Cao-Berg

HPV and Cervical Cancer Research Assistant

San Juan, Puerto Rico

UPR MEDICAL SCIENCE / UAGMA GURABO

May 2025 - August 2025

- Studied HPV relation to cervical cancer using machine learning approaches
- Organized data and created patient cohorts
- Conducted feature importance analysis
- **Mentors:** Dr. Roche-Lima, Dr. Tosado

COVID-19 Diabetes Research Assistant

San Juan, Puerto Rico

- Evaluated COVID patients with diabetes using ML models
- Organized data and created cohorts
- Analyzed feature importance
- **Mentors:** Dr. Roche-Lima, Dra. Lluberes

Brain Image Library Dashboard Developer

CARNEGIE MELLON UNIVERSITY

Pittsburgh, Pennsylvania

June 2023 - August 2023

- Designed dashboard requirements and metrics
- Used Bridges 2 for metadata evaluation and separation
- Implemented metrics in the dashboard system
- **Mentor:** I. Cao-Berg

Bee Monitoring Interface Developer

UPR RIO PIEDRAS

San Juan, Puerto Rico

August 2023 - May 2024

- Added new features to platform for live bee monitoring
- Worked with Python, JavaScript, and MongoDB
- Learned API development techniques
- **Mentor:** Dr. Megret

Kidney Disease Reference Intervals Researcher

UPR MEDICAL SCIENCE

San Juan, Puerto Rico

April 2023 - June 2023

- Developed unsupervised ML method for chronic kidney disease reference intervals
- **Published:** Velez, J., LeBien, J. & Roche-Lima, A. Sci Rep 13, 17198 (2023)
- **Mentor:** Dr. Roche-Lima

Ethnic Disparities Research Assistant

UPR MEDICAL SCIENCE / UPR RIO PIEDRAS

San Juan, Puerto Rico

September 2022 - May 2023

- Researched ethnic disparities in urinary and serum biomarkers
- Applied ML methods for lab test reference intervals by race and ethnicity
- Worked with Python and All of Us dataset
- **Mentors:** Dr. Roche-Lima, Dr. H. Ortiz

Skills

Programming Languages Python (intermediate), C++ (intermediate), C (intermediate), Bash (intermediate), JavaScript (basic), C# (basic), Visual Basic (basic)

Databases & Tools MongoDB, Grafana, Bridges 2, Git

Spoken Languages Spanish (native), English (native)

Teaching Experience

AI/ML Course Assistant

UPR MEDICAL SCIENCE

San Juan, Puerto Rico

February 2023 - May 2023

- Assisted with “Artificial Intelligence and Machine Learning applied to Health Disparities Research (Version 2)”
- Expanded course to professionals outside UPR system
- Supervised student projects and answered technical questions
- Applied student feedback to improve course materials
- **Mentor:** Dr. Roche-Lima

AI/ML Course Developer

UPR MEDICAL SCIENCE

San Juan, Puerto Rico

April 2022 - July 2022

- Developed pilot course “AIML + HDR” for graduate students and professionals
- Created PowerPoints, example exercises, and instructional videos
- Graded assignments and provided student support
- **Published:** Heredia-Negron, Frances, et al. Int. J. Environ. Res. Public Health 20.3 (2023): 2726
- **Mentor:** Dr. Roche-Lima

STEM Instructor

C3TECH

- Taught STEM concepts to elementary students
- Focused on programming with Scratch and computational logic

Caguas, Puerto Rico

June - July 2019

Awards and Fellowships

HuBMAP Underrepresented Student Internship Extension

PITTSBURGH SUPERCOMPUTING CENTER

- Continuation of HuBMAP research project
- **Mentors:** Ivan Cao-Berg, Dr. H. Ortiz

Pittsburgh, Pennsylvania

November 2024 - Present

REU - SSP HuBMAP Project

CARNEGIE MELLON UNIVERSITY

Pittsburgh, Pennsylvania

May 2024 - August 2024

REU - SSP Brain Image Library

CARNEGIE MELLON UNIVERSITY

Pittsburgh, Pennsylvania

June 2023 - August 2023

PRIME Fellowship

ETHNIC DISPARITIES RESEARCH

- Machine learning methods for reference intervals by race and ethnicity
- **Mentors:** Dr. Roche-Lima, Dr. H. Ortiz

Online

September 2022 - May 2023

Publications

Published

1. Velev, J., LeBien, J. & Roche-Lima, A. Unsupervised machine learning method for indirect estimation of reference intervals for chronic kidney disease in the Puerto Rican population. *Scientific Reports* 13, 17198 (2023).
2. Heredia-Negron, Frances, et al. "Evaluation of AIML+ HDR—A Course to Enhance Data Science Workforce Capacity for Hispanic Biomedical Researchers." *International Journal of Environmental Research and Public Health* 20.3 (2023): 2726.

Under Review

- Validating Gaussian Mixture Models for Accurate Health Interval Determination: A Multi-Ethnic Analysis of Laboratory Parameters from Electronic Health Records

Conference Presentations

- SACNAS Chapter UPR-RP (March 2026, San Juan, PR) - HuBMAP Visualization Tool for FAIR Metrics
- ABRCMS (November 2024, Pittsburgh, PA) - HuBMAP Dashboard Development (*Accepted*)
- SACNAS (October 2024, Arizona) - HuBMAP Dashboard Development
- Gateway Conference (October 2024, Montana) - HuBMAP Dashboard Development (*Accepted*)
- SAIL Conference (May 2024, Rio Grande, PR) - Health Disparities Reference Intervals
- SIDIM (March 2024, Humacao, PR) - Bee Monitoring Web Application
- Mini-Symposium AI Applied to Health Disparities (January 2024, San Juan, PR) - Health Disparities Research
- SACNAS (October 2023, Oregon) - Brain Image Library Dashboard
- AIM-AHEAD Conference (August 2023, Bethesda, WA) - Ethnic Disparities in Biomarkers
- CIO & IT Leader Conference (September 2022, San Juan, PR) - Student programmer presentation

Certifications

VADSTI Data Science Training

DATA SCIENCE & AI/ML CERTIFICATION

AIM-AHEAD

February 2023